Toxoplasmosis [protozoan disease]

Microbiology III

- Toxoplasmosis is a zoonosis type of infection caused by the protozoan ***Toxoplasma gondii***.
- Humans are infected by **ingesting oocysts** excreted by the definitive hosts (**cats**) or by eating **unprocessed meat** containing *Toxoplasma* cysts.
- If a women contracts toxoplasmosis for the first time during **pregnancy** it affects the child with severe defects.

- **Occurrence:** *T. gondii* occurs worldwide. It infects wide spectrum of warm blooded vertebrates such as humans, sheep, pigs, cattle, horses, dogs, cats, wild mammals, bird species

Causative agent: *Toxoplasma gondii*

- Various strains of *T. gondii* differ in virulence and certain biological and genetic characteristics.
- The life cycle of *T. gondii* includes various stages:

1. Tachyzoites
2. Bradyzoites
3. Oocysts

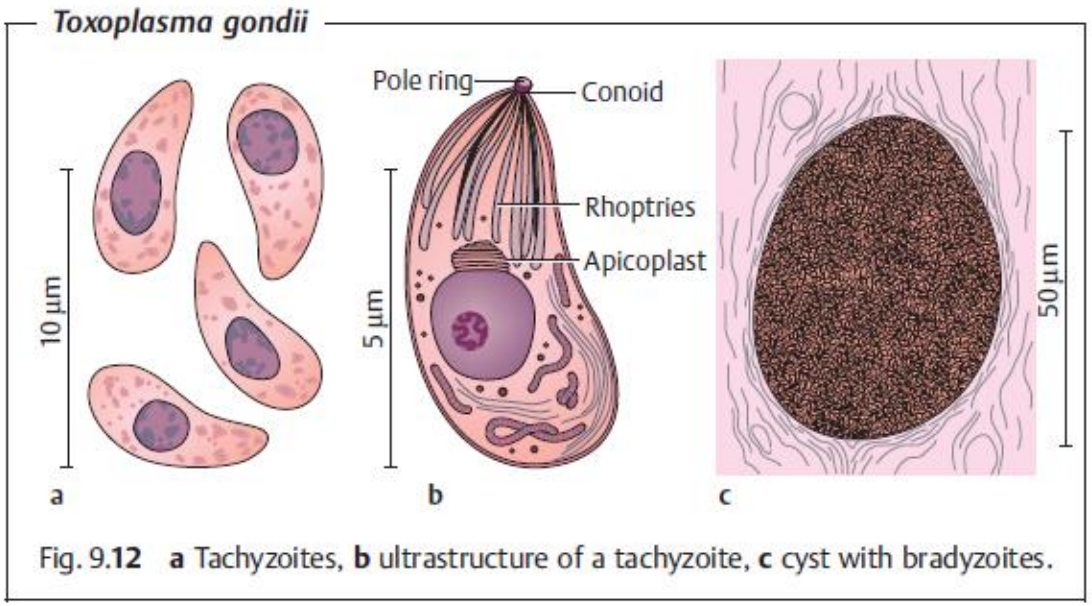
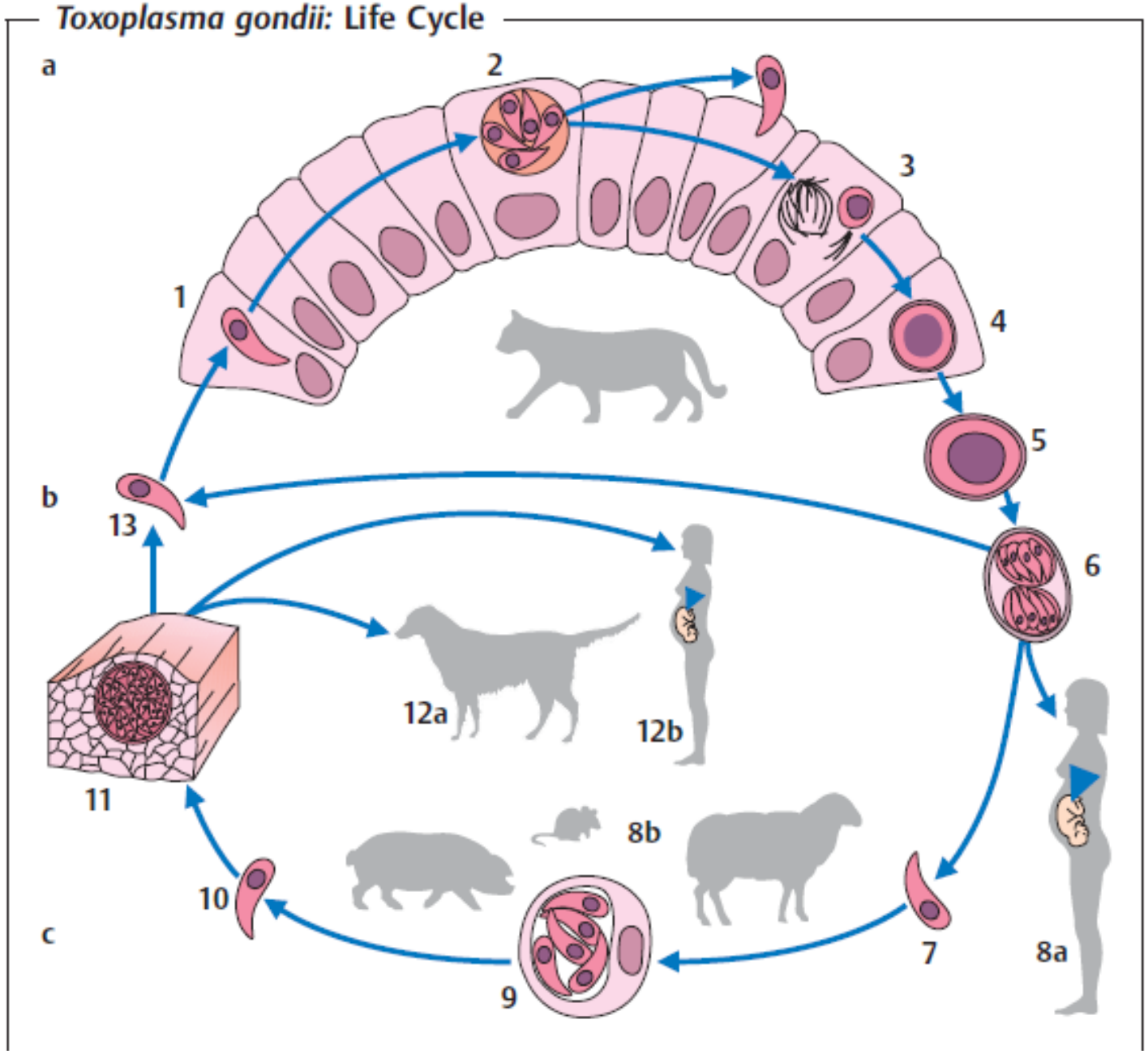


Fig. 9.12 **a** Tachyzoites, **b** ultrastructure of a tachyzoite, **c** cyst with bradyzoites.

- **Tachyzoites** (endozoites):
- are proliferative forms that reproduce within the host cells by endogenous budding of the mother cell.
- Tachyzoites are sickle shaped, 4–7 µm long and 2–4 mm wide
- Tachyzoites also multiply in experimental animals and cell cultures.
- This stage is highly labile outside of a host and usually does not survive the stomach passage following ingestion.

- **Bradyzoites** (cystozoites) are stages produced by slow reproduction within the cysts (4–8 X 2–4 µm).
- The cysts develop intracellularly in various tissues
- Have a relatively resistant wall, grow as large as 150 µm,
- and can contain up to several thousand bradyzoites
- They have a long lifespan in the host.
- Humans and animals can be infected by ingestion of meat containing cysts

- **Oocysts** are round and encysted stages of the organism
- surrounded by a resistant cyst wall, and are approx. 9 X 14 µm in size.
- They are the final product of a sexual reproductive cycle in the intestinal epithelia of Cats.
- They contain a zygote and are shed in feces.
- Sporulation takes place within two to four days, producing two sporocysts with four sporozoites each.
- Sporulated oocysts are infective for humans and animals.


Toxoplasma gondii: Life Cycle
a
b
c
1
2
3
4
5
6
7
8a
8b
9
10
11
12a
12b
13

a Development in the definitive host (cat): intestinal phase with production of sexual forms:

1 Toxoplasma that has penetrated into a small intestine epithelial cell;

2 asexual reproductive stage with merozoites (can continue for several generations). The arrow indicates that an extra intestinal cycle precedes the enteral cycle when a cat is infected with oocysts;

3 production of sexual forms (gamogony) and formation of the zygote;

4 oocyst.

b External phase with sporogony:

5 unsporulated oocyst, shed in cat feces;

6 sporulated oocyst with two sporocysts each containing four sporozoites.

c Development in an intermediate host (mammals, birds, humans): extra intestinal phase with only asexual multiplication of parasites;

7 sporozoite released in organism following oral ingestion of oocysts;

8a human infection; **8b** infection of various animal species;

9 asexual reproductive stages (tachyzoites) in a somatic cell; **10** free tachyzoite;

11 cyst with bradyzoites in musculature;

12a infection of a dog with Toxoplasma cysts in animal meat; **12b** infection of a human with Toxoplasma cysts;

13 infection of cat with infective stages of cysts in meat or with oocysts.

Life cycle:

- Intestinal phase
- External phase
- Extra intestinal phase

Intestinal phase:

- Production of sexual forms (gamogony) takes only place in enterocytes of definitive hosts - cats.
- Primary infection of a cat with *Toxoplasma* cysts in raw meat
- Asexual reproductive forms at first develop in the **small intestine** epithelium
- followed by sexually differentiated stages and **oocysts**
- Oocysts are shed in feces (3-9 days)
- Sporulated oocysts - 20–35 days

- **External phase:**
- Oocysts excreted in cat feces sporulate at room temperature within 2-4 days, become infective.
- Kept moist, they remain infective for up to 5 years
- And not killed by standard disinfectant agents.
- They die within a few minutes at temperatures exceeding 55°C.

Extra intestinal phase:

- This phase follows a peroral infection with oocysts or cysts
- Observed in:
  - intermediate hosts (dogs, sheep, pigs, other vertebrates, birds)
  - dead-end hosts (humans)
  - definitive hosts (cats)

- Starting from the intestine, the *Toxoplasma* organisms travel in blood or lymph to various organs and multiply in nucleate host cells.
- Repeated cycles produces 32 daughter cells that expand the host cells before it bursts.
- The released tachyzoites attack neighboring cells.
- This results in focal necroses and inflammatory reactions in affected tissues.
- Infects the fetus from the mother in the placenta
- Cysts remain in tissues viable for many years without causing noticeable damage to the host.

Pathogenicity and clinical manifestations.:

- Focal necrotic, inflammatory and immunopathological processes are the basis of the pathogenesis
- Varied clinical manifestations observed in toxoplasmosis.
- Cases are differentiated as:
  - postnatal
  - prenatal infections

Postnatal Toxoplasma Infection:

- Primary infection in immunocompetent persons: This is the most frequent form without clinical manifestations.
- Primary infection during pregnancy. This may cause prenatal infection of the fetus and thus become a significant threat
- Primary infection in immunosuppressed persons.
- Reactivation toxoplasmosis in cases of immune deficiency.

Prenatal Toxoplasmosis:

- Occurs in mothers during 1st pregnancy and risk of infection to the child.
- Causes brain and eye damages.

Diagnosis:

- Toxoplasmosis is normally diagnosed serologically by detection of parasite-specific IgG and IgM antibodies.
- Prenatal diagnostics: PCR for the detection of *Toxoplasma* DNA indicates fetal infection.
- In suspected cases, the blood or cerebrospinal fluid should by examined using the PCR.

Treatment:

- Chemotherapy for pregnancy– spiramycin
- Immunosuppressed patients – pyrimethamine/sulfadiazine or pyrimethamine/clindamycin.

Prevention:

- Eating meat that has been thoroughly heated (70C) or deep-frozen.
- Close contact with cats should be avoided.